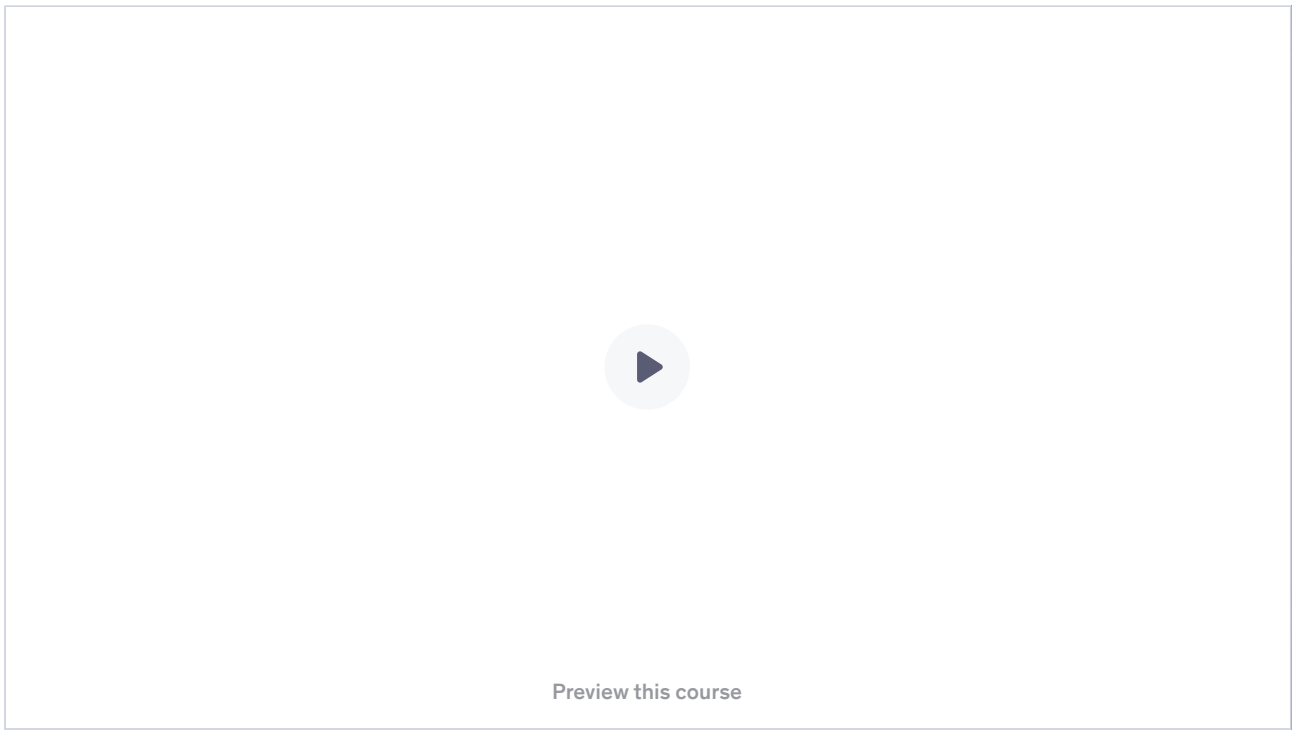
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- ✓ Project 1: Make AI-powered brochure generator that scrapes and navigates company websites intelligently.
- ✓ Project 2: Build Multi-modal customer support agent for an airline with UI and function-calling.
- ✓ Project 3: Develop Tool that creates meeting minutes and action items from audio using both open- and closed-source models.
- ✓ Project 4: Make AI that converts Python code to optimized C++, boosting performance by 60,000x!
- ✓ Project 5: Build AI knowledge-worker using RAG to become an expert on all company-related matters.
- ✓ Project 6: Capstone Part A – Predict product prices from short descriptions using Frontier models.
- ✓ Project 7: Capstone Part B – Execute Fine-tuned open-source model to compete with Frontier in price prediction.
- ✓ Project 8: Capstone Part C – Build Autonomous multi agent system collaborating with models to spot deals and notify you of special bargains.

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- ✓ Compare and contrast the latest techniques for improving the performance of your LLM solution, such as RAG, fine-tuning and agentic workflows
- ✓ Weigh up the leading 10 frontier and 10 open-source LLMs, and be able to select the best choice for a given task

Explore related topics

Large Language Models (LLM) | Artificial Intelligence (AI) | ChatGPT | Data Science | Development

This course includes:

- 33.5 hours on-demand video
- Access on mobile and TV
- 1 article
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Course content

8 sections • 210 lectures • 33h 27m total length

Expand all sections

Week 1 - Build Your First LLM Product: Exploring Top Models — 37 lectures • 5hr 46min

Day 1 - Running Your First LLM Locally with Ollama and Open Source Models	Preview	10:44
Day 1 - Spanish Tutor Demo with Open-Source Models & Course Overview		8:54
Your Path to Becoming a Proficient AI Engineer		3:42
Day 1 - Setting Up Your LLM Development Environment with Cursor and UV		5:54
Day 1 - Setting Up Your PC Development Environment with Git and Cursor		9:50
Day 1 - Mac Setup: Installing Git, Cloning the Repo, and Cursor IDE		9:47
Day 1 - Installing UV and Setting Up Your Cursor Development Environment		7:52
Day 1 - Setting Up Your OpenAI API Key and Environment Variables		12:28
Day 1 - Installing Cursor Extensions and Setting Up Your Jupyter Notebook		9:04
Day 1 - Running Your First OpenAI API Call and System vs User Prompts		11:41

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Day 2 - LLM Engineering Building Blocks: Models, Tools & Techniques	9:22
Day 2 - Your 8-Week Journey: From Chat Completions API to LLM Engineer	2:51
Day 2 - Frontier Models: OpenAI GPT, Claude, Gemini & Grok Compared	6:14
Day 2 - Open-Source LLMs: LLaMA, Mistral, DeepSeek, and Ollama	12:03
Day 2 - Chat Completions API: HTTP Endpoints vs OpenAI Python Client	10:11
Day 2 - Using the OpenAI Python Client with Multiple LLM Providers	7:41
Day 2 - Running Ollama Locally with OpenAI-Compatible Endpoints	10:28
Day 3 - Base, Chat, and Reasoning Models: Understanding LLM Types	10:44
Day 3 - Frontier Models: GPT, Claude, Gemini & Their Strengths and Pitfalls	12:56
Day 3 - Testing ChatGPT-5 and Frontier LLMs Through the Web UI	9:43
Day 3 - Testing Claude, Gemini, Grok & DeepSeek with ChatGPT Deep Research	11:33
Day 3 - Agentic AI in Action: Deep Research, Claude Code, and Agent Mode	11:22
Day 3 - Frontier Models Showdown: Building an LLM Competition Game	10:14
Day 4 - Understanding Transformers: The Architecture Behind GPT and LLMs	12:46
Day 4 - From LSTMs to Transformers: Attention, Emergent Intelligence & Agentic A	9:08
Day 4 - Parameters: From Millions to Trillions in GPT, LLaMA & DeepSeek	8:26
Day 4 - What Are Tokens? From Characters to GPT's Tokenizer	4:02
Day 4 - Understanding Tokenization: How GPT Breaks Down Text into Tokens	8:13
Day 4 - Tokenizing with tiktoken and Understanding the Illusion of Memory	10:56
Day 4 - Context Windows, API Costs, and Token Limits in LLMs	10:49
Day 5 - Building a Sales Brochure Generator with OpenAI Chat Completions API	9:03
Day 5 - Building JSON Prompts and Using OpenAI's Chat Completions API	10:43
Day 5 - Chaining GPT Calls: Building an AI Company Brochure Generator	9:07
Day 5 - Building a Brochure Generator with GPT-4 and Streaming Results	11:17
Day 5 - Business Applications, Challenges & Building Your AI Tutor	9:56
Week 2 - Build a Multi-Modal Chatbot: LLMs, Gradio UI, and Agents	24 lectures • 3hr 41min
Week 3 - Open-Source Gen AI: Automated Solutions with HuggingFace	23 lectures • 3hr 36min
Week 4 - LLM Showdown: Evaluating Models for Code Gen & Business Tasks	21 lectures • 3hr 9min
Week 5 - Mastering RAG: Build Advanced Solutions with Vector Embeddings	32 lectures • 5hr 25min

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Week 7 - Fine-tuned open-source model to compete with Frontier model	24 lectures • 4hr
Week 8 - Build Autonomous multi agent system	22 lectures • 3hr 6min

Requirements

- Familiarity with Python. This course will not cover Python basics and is completed in Python.
- A PC with an internet connection is required. Either Mac (Linux) or Windows.
- We recommend that you allocate around $5 for API costs to work with frontier models. However, you can complete the course using open-source models if you prefer.

Description

Mastering Generative AI and LLMs: An 8-Week Hands-On Journey

Accelerate your career in AI with practical, real-world projects led by industry veteran Ed Donner. Build advanced Generative AI products, experiment with over 20 groundbreaking models, and master state-of-the-art techniques like RAG, QLoRA, and Agents.

What you'll learn

• **Build advanced Generative AI products** using cutting-edge models and frameworks.

• **Experiment with over 20 groundbreaking AI models**, including Frontier and Open-Source models.

• **Develop proficiency with platforms** like HuggingFace, LangChain, and Gradio.

• **Implement state-of-the-art techniques** such as RAG (Retrieval-Augmented Generation), QLoRA fine-tuning, and Agents.

• **Create real-world AI applications**, including:

• A multi-modal customer support assistant that interacts with text, sound, and images.

• An AI knowledge worker that can answer any question about a company based on its shared drive.

• An AI programmer that optimizes software, achieving performance improvements of over 60,000 times.

• An ecommerce application that accurately predicts prices of unseen products.

• **Transition from inference to training**, fine-tuning both Frontier and Open-Source models.

• **Deploy AI products to production** with polished user interfaces and advanced capabilities.

• **Level up your AI and LLM engineering skills** to be at the forefront of the industry.

About the Instructor

I'm **Ed Donner**, an entrepreneur and leader in AI and technology with over 20 years of experience. I've co-founded and sold my own AI startup, started a second one, and led teams in top-tier financial institutions and startups around the world. I'm passionate about bringing others into this exciting field and helping them become experts at the forefront of the industry.

Projects:

Project 1: AI-powered brochure generator that scrapes and navigates company websites intelligently.

Project 2: Multi-modal customer support agent for an airline with UI and function-calling.

Project 3: Tool that creates meeting minutes and action items from audio using both open- and closed-source models.

Project 4: AI that converts Python code to optimized C++, boosting performance by 60,000x!

Project 5: AI knowledge-worker using RAG to become an expert on all company-related matters.

Project 6: Capstone Part A – Predict product prices from short descriptions using Frontier models

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Project 8: Capstone Part C – Autonomous agent system collaborating with models to spot deals and notify you of special bargains.

Why This Course?

• Hands-On Learning: The best way to learn is by doing. You'll engage in practical exercises, building real-world AI applications that deliver stunning results.

• Cutting-Edge Techniques: Stay ahead of the curve by learning the latest frameworks and techniques, including RAG, QLoRA, and Agents.

• Accessible Content: Designed for learners at all levels. Step-by-step instructions, practical exercises, cheat sheets, and plenty of resources are provided.

• No Advanced Math Required: The course focuses on practical application. No calculus or linear algebra is needed to master LLM engineering.

Course Structure

Week 1: Foundations and First Projects

• Dive into the fundamentals of Transformers.

• Experiment with six leading Frontier Models.

• Build your first business Gen AI product that scrapes the web, makes decisions, and creates formatted sales brochures.

Week 2: Frontier APIs and Customer Service Chatbots

• Explore Frontier APIs and interact with three leading models.

• Develop a customer service chatbot with a sharp UI that can interact with text, images, audio, and utilize tools or agents.

Week 3: Embracing Open-Source Models

• Discover the world of Open-Source models using HuggingFace.

• Tackle 10 common Gen AI use cases, from translation to image generation.

• Build a product to generate meeting minutes and action items from recordings.

Week 4: LLM Selection and Code Generation

• Understand the differences between LLMs and how to select the best one for your business tasks.

• Use LLMs to generate code and build a product that translates code from Python to C++, achieving performance improvements of over 60,000 times.

Week 5: Retrieval-Augmented Generation (RAG)

• Master RAG to improve the accuracy of your solutions.

• Become proficient with vector embeddings and explore vectors in popular open-source vector datastores.

• Build a full business solution similar to real products on the market today.

Week 6: Transitioning to Training

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Week 7: Advanced Training Techniques

- Dive into advanced training techniques like **QLoRA fine-tuning**.
- Train an **open-source model** to outperform Frontier models for specific tasks.
- Tackle challenging projects that push your skills to the next level.

Week 8: Deployment and Finalization

- Deploy your commercial product to production with a polished **UI**.
- Enhance capabilities using **Agents**.
- Deliver your first **productionized, agentized, fine-tuned LLM model**.
- Celebrate your mastery of AI and LLM engineering, ready for a new phase in your career.

Who this course is for:

- Aspiring AI engineers and data scientists eager to break into the field of Generative AI and LLMs.
- Professionals looking to upskill and stay competitive in the rapidly evolving AI landscape.
- Developers interested in building advanced AI applications with practical, hands-on experience.
- Individuals seeking a career transition or aiming to enhance productivity through LLM-built frameworks.

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Where this leads: roles that ship. **AI Engineer** and **LLM Engineer** for those who love building; platform and MLOps paths for those drawn to reliability at scale; product and leadership tracks for the people moving **Agentic AI** from slide decks to business outcomes. The through-line is the same: learn fast, build faster, measure everything, iterate.

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Ed Donner

Ed Donner



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Ed Donner is a technology leader and repeat founder of AI startups. He's the co-founder and CTO of Nebula, the platform to source, understand, engage and manage talent, using Generative AI and other forms of machine learning. Nebula's long-term goal is to help people discover their potential and pursue their reason for being.

Previously, Ed was the founder and CEO of AI startup untapt, an Accenture Fintech Innovation Lab company, acquired in 2021. Before that, Ed was a Managing Director at JPMorgan Chase, leading a team of 300 software engineers in Risk Technology across 3 continents, after a 15-year technology career on Wall Street. Ed holds a patent for a Deep Learning matching engine issued in 2023, and an MA in Physics from Oxford.

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★ 4.7 course rating • 36K ratings

David

★★★★★ 12 days ago

Yes, yes, I skipped to Week 5 almost immediately. I've watched some of the videos 3-4 times, most of the course twice and the first and eighth week only once. Incredibly practical and the best approach to explaining traditional ML models (Dense, Sequential) I've ever seen. My Python skills suck, but I was able to build my own RAG and other tools from the instruction alone. My only complaint is the capstone

Leo Budiharto

★★★★½ 25 days ago

I wanted to give it 5 stars, but some sections of this course to be honest: felt like they were too far for beginners (such as jumping straight into advanced topics and too many detailed coding jumping around from Cursor IDE to Google colab, from jupyter notebook ipynb source back and forth to python (.py) source code and vice versa that aren't intended for beginners, in sections 7 and 8)

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Alankar

1 month ago

Mr. Donner is exceptionally enthusiastic and encouraging, on top of being extremely knowledgeable. So, taking this course was a great experience. It should provide a lot of people with a pretty solid foundation in many aspects of AI.

Helpful?

Ilhan

1 month ago

Awesome course!!! The content was highly informative and well-structured. Ed Donner has an excellent way of explaining complex topics in a clear and engaging manner. I really enjoyed taking this course and found it both valuable and rewarding.

Helpful?

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